

Week 2 Report: Literature Review and Dataset Collection

Project: Determining the Role of Big Data Applications in Fraud Detection and Financial Security in Financial Institutions of Ireland

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Module: Open Data Practice, Research Practicum

1. What this week covers

My Gantt chart has the literature review running from Week 1 to Week 3 and dataset collection in Weeks 2 and 3. So this week I did two things. I compared the fifteen papers from Week 1 with each other to see where they agree and what they miss, and I collected and checked the dataset. Data preprocessing starts next week and full exploratory analysis is planned for Week 4, so the data checks here are only the ones I need to plan that work.

2. Literature review progress

Looking across the papers, Random Forest and other tree ensembles are the models that keep doing best. Albalawi and Dardouri (2025) and Andrade-Arenas and Yactayo-Arias (2025) both found Random Forest had the best F1 on the Kaggle European card data (0.8256 and 0.872). Preciado Martinez et al. (2025) found the same on 565,000 real bank transfers, and the review by Baisholan et al. (2025) says tree ensembles are the most common choice across the field. Jamil et al. (2025) and Salunke et al. (2025) compare the same three models I plan to use, Logistic Regression, Decision Tree and Random Forest, and Random Forest or a combination of the three comes out ahead.

The second point is that accuracy does not tell the full story. Andrade-Arenas and Yactayo-Arias (2025) report accuracy of 0.999 for several models, yet Logistic Regression only managed an F1 of 0.110. Haider et al. (2024) and Alrasheedi (2025) report mainly accuracy, which makes it hard to know how many frauds were caught. Baisholan et al. (2025) point out that AUC-PR is used much less than it should be. I will therefore report precision, recall, F1 and AUC-PR, not just accuracy.

Third, SMOTE is used in many papers, but it is not always a clear win. Ma et al. (2026) found that resampling is not always helpful and that it should be judged on a test set that keeps the real class balance. Most other papers do not say clearly whether they applied SMOTE before or after splitting the data. If it is applied before the split, synthetic copies of test frauds end up in training and the results look better than they are. I will only apply it to the training data and compare it with class weights.

Fourth, almost every experimental paper uses the same Kaggle dataset. This makes results easy to compare, but none of them mention the duplicate rows I found in the file (Section 3). The features are also PCA components, so the models are hard to explain to a bank or a regulator.

Finally, the regulatory side is thin. Horobets et al. (2025) show that EU rules on anti-money laundering and the AI Act ask for explainability and fairness that current AI systems do not fully deliver. Gkegkas et al. (2025) and Bian (2025) also call for more explainable models. Jamil et al. (2025) is set in the USA and Khan et al. (2026) in Pakistan. None of the papers look at Irish institutions, which is the gap my project focuses on.

Table 1 summarises the experimental papers I will compare my results against.

Paper	Data	Models	Main result
Albalawi and Dardouri (2025)	Kaggle European cards, PaySim	LR, DT, RF, XGBoost, deep learning	RF best: F1 0.8256, ROC-AUC 0.9759
Andrade-Arenas and Yactayo-Arias (2025)	Kaggle European cards	Seven models with SMOTE	RF F1 0.872, AUC 0.978; LR F1 0.110
Ma et al. (2026)	Kaggle European cards	RF feature selection, tuned XGBoost	PR-AUC 0.798, fraud F1 0.809
Preciado Martinez et al. (2025)	565,000 bank transfers	RF, neural network, Naive Bayes	RF 95.79% fraud detection
Jamil et al. (2025)	US transaction data	LR, DT, RF	RF fewest errors
Salunke et al. (2025)	Public card dataset	LR, DT, RF, hybrid	Hybrid ensemble best
Haider et al. (2024)	Imbalanced card dataset	Six models with PCA and SMOTE	About 96 to 97% accuracy
Alrasheedi (2025)	Three datasets	Seven models	RF 0.97 accuracy on imbalanced data

Table 1. Experimental papers used for comparison.

From this I have narrowed my research gaps to five points: SMOTE timing is often not reported, accuracy is overused, duplicates in the benchmark are ignored, the PCA features are hard to explain, and there is no Irish context. Each one links to a step in my method.

3. Dataset collection

Before choosing a dataset I wrote down what I needed: labelled card or payment transactions, an open licence, European data if possible, no personal information, and a dataset used by the papers I compare with. I looked at four options. IEEE-CIS has many raw identity fields and a competition-only licence. PaySim (Lopez-Rojas et al., 2016) is simulated mobile money. The Bank Account Fraud dataset (Jesus et al., 2022) is about account applications. The Kaggle Credit Card Fraud Detection dataset from ULB meets all five points and is the one named in my proposal, so I chose it.

For the Irish context I also collected the Central Bank of Ireland payment fraud statistics for 2022 to 2024. These are summary figures, not training data. They show 160 million euro of fraudulent payments and about 815,000 fraudulent transactions in 2024 (Central Bank of Ireland, 2025).

I downloaded the dataset with the Kaggle API. Kaggle lists the licence as DbCL-1.0. The file is creditcard.csv (150.8 MB). I saved its SHA-256 checksum so I can show later that I used the same file: 76274b691b16a6c49d3f159c883398e03ccd6d1ee12d9d8ee38f4b4b98551a89

The data owners replaced the original features with 28 PCA components, V1 to V28, to protect cardholders. Only Time and Amount keep their original meaning. Table 2 describes the columns.

Column	Type	Range	Meaning
Time	float64	0.0 to 172792.0	Seconds since the first transaction in the file
Amount	float64	0.0 to 25691.16	Transaction amount (EUR, not scaled)
Class	int64	0.0 to 1.0	Target label: 1 = fraud, 0 = legitimate
V1 to V28	float64	-113.74 to 120.59	PCA components, already centred on zero

Table 2. Data dictionary (the full 31 columns are in data_dictionary.csv).

The file has 284,807 rows and 31 columns with no missing values. It covers 48 hours, which is two days of transactions. I found 1,081 rows that are exact copies of other rows, and 19 of them are fraud. If these stay in, the same transaction can end up in both the training and the test set, so I will remove them before splitting. That leaves 283,253 legitimate and 473 fraud transactions. There are also 1,825 transactions with an amount of zero, which I will keep because small or zero charges are often used to test a stolen card.

Only 492 of the 284,807 transactions (0.173%) are fraud, about one in 578. A model that always says legitimate would score 99.83% accuracy, which is exactly why accuracy is not a good measure here. The train and test split has to be stratified so both sets keep some fraud cases.

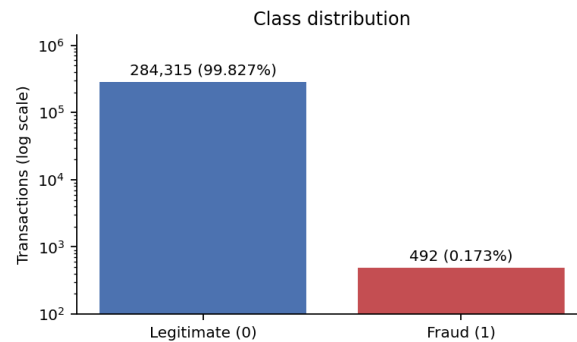


Figure 1. Legitimate and fraud transactions (log scale).

A first look at the data shows a few useful patterns. The median fraud amount is 9.25, lower than 22.0 for legitimate transactions, but the average fraud is higher because of some large ones (Figure 2). Fraud is also more common at night: 0.518% of transactions between midnight and 6am are fraud compared with 0.141% for the rest of the day (Figure 3), so an hour of day feature may help. The components most linked to fraud are V17, V14, V12 and V10 (Figure 4).

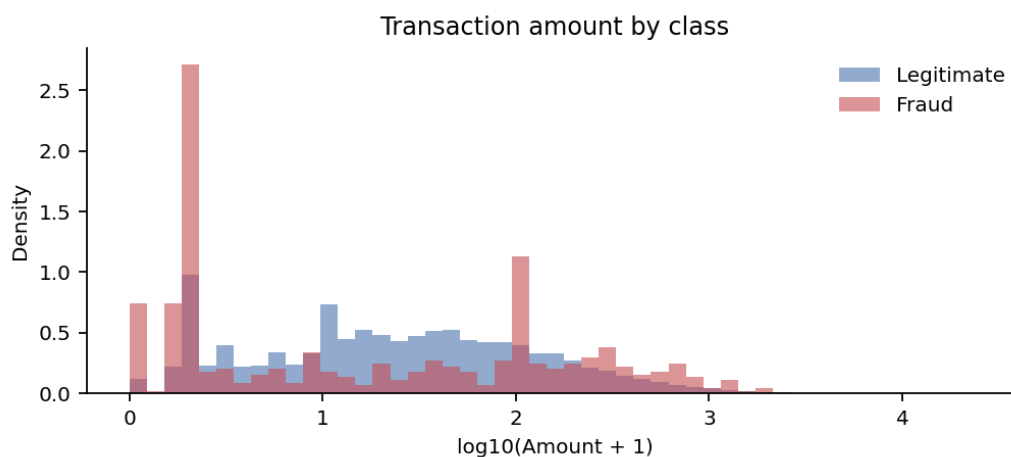


Figure 2. Transaction amounts by class on a log scale.

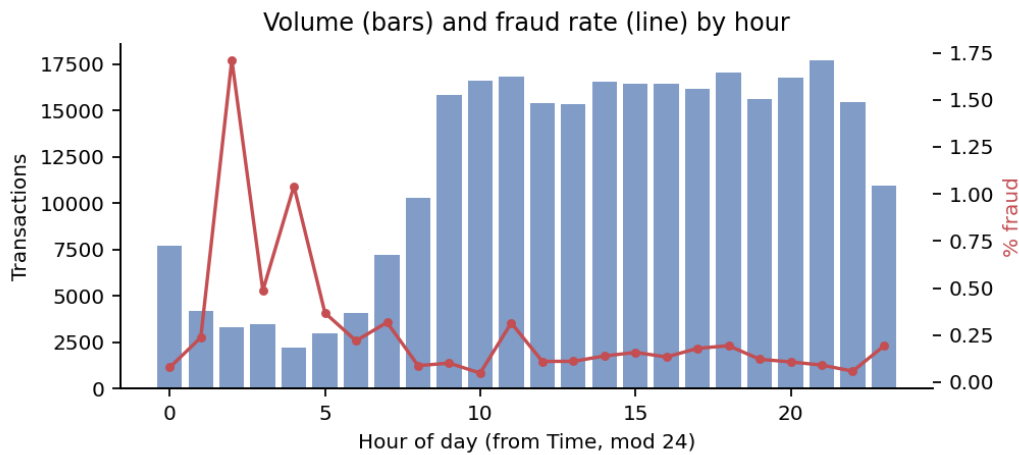


Figure 3. Transactions per hour (bars) and fraud rate (line).

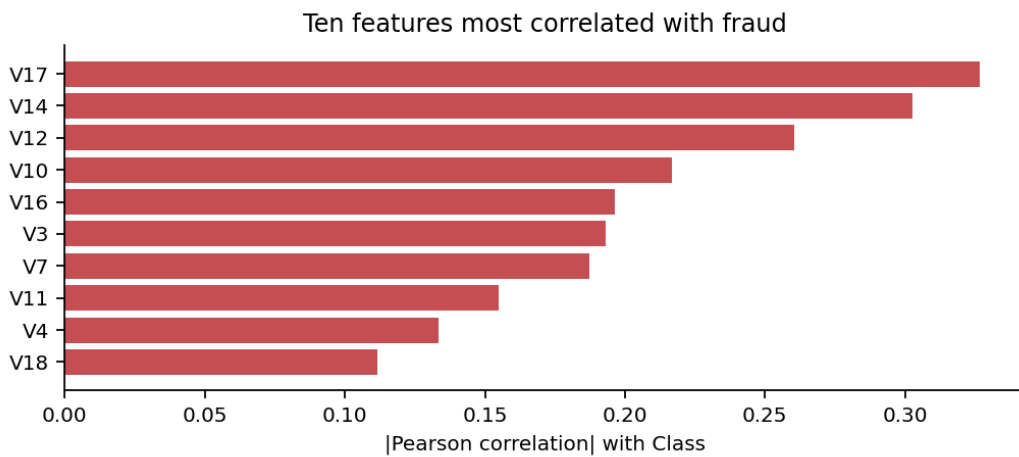


Figure 4. Features most correlated with fraud.

4. Ethics

The dataset has no names, card numbers or other personal details. Only PCA components, Time and Amount are released, so no one can be identified from it. I use it for academic research under its licence, keep it on my own machine and do not share it. The Central Bank figures are public. Because the data is from European cardholders in 2013 and not from Irish banks, I will present my results as a test of the methods and base the Irish recommendations on the literature and Central Bank data.

5. Next week

In Week 3 I will finish the literature review with a few more papers on SMOTE and evaluation, and on EU rules for automated fraud decisions. I will also start preprocessing: removing the duplicates, scaling Time and Amount, and creating one fixed stratified train and test split that I will use for every model.

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